

First R (Base) Code Quiz – MTCARS

Instructions: Create a script file using base R code only to read-in and modify the mtcars file. Submit your R script file via Blackboard. As always, include comments in your code to explain all of the major steps.

1. Read-in the mtcars file that is a preloaded practice dataset using R Studio.
2. Create a copy of the original data frame named "mtcars2," and create a new variable "car_name" from the rowname in this new data frame.
3. Create a numeric ID variable using the rownames function.
4. Drop the rowname column by using the rownames and <- NULL functions.
5. Use the summary function to find the first and third quartile and mean and median of the "wt" (weight) variable.
6. Create a new variable (new_weight) that classifies weight into three levels ("light", "medium", "heavy") based on the first and third quartile weights (from summary) to define the three categories.
7. Perform a frequency count of the new_weight variable and report the count via a comment.
8. Add a new car to the dataframe by creating a new row (vector) of data that includes the following data:

Car_name	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Chrysler_300	21	6	111	175	4.11	2.9	18.50	1	1	4	1

9. Sort all cars by mpg and report the number of cars with gas mileage higher than 19 mpg.
10. Create a dataframe named "good_mpg" that includes only vehicles with gas mileage above 20 mpg.
11. Submit your R.script via Blackboard with comments including your name.